

Chapter 3 - Curve Sketching

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#Curve Sketching

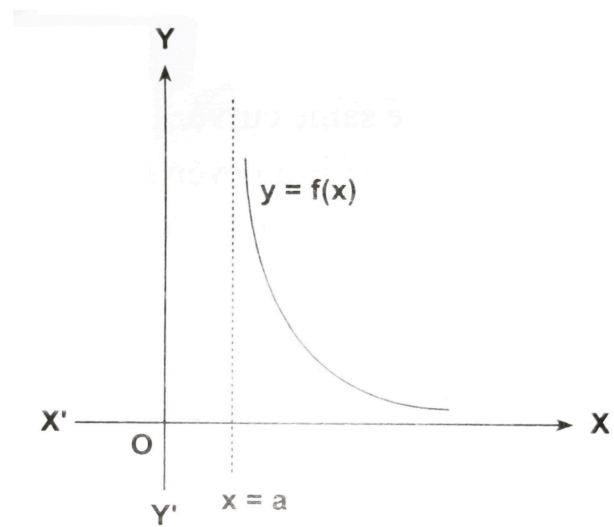
Curve sketching is the representation of the given function in its graphical form. It is worked out with the help of the different characteristics which the graph satisfies.

Terminology

Period: If $f(x) = f(x + k)$ for all $x \in \text{Dom}(f)$ then the number k is said to be its period.

For example, $\sin(x + 2\pi) = \sin x \Rightarrow 2\pi$ is the period of function $y = \sin x$

Asymptote: An asymptote is a line which gets closer and closer to a curve but never meets it.



Key Steps before Sketching a Function

1. **Origin:** Check whether the function contains origin (i.e. $x = 0 \Rightarrow y = 0$) or not
2. **Points/Intercepts on Axes:** See where the locus (curve) cuts the axes
 - a. **X-intercept:** The value of x when $y = 0$
 - b. **Y-intercept:** The value of y when $x = 0$
3. **Symmetry:** Determine the axis of symmetry of the function
 - a. **Symmetric about Y-axis:** $f(-x) = f(x)$
 - b. **Symmetric about X-axis:** $f(-y) = -f(y)$
 - c. **Symmetric about origin:** $f(-x) = -f(x)$

#Even and Odd Functions

Even function	If $f(-x) = f(x)$ for all $x \in \text{Dom}(f)$ then the function $f(x)$ is said to be an even function. Such functions are symmetric about the Y-axis.
Odd function	If $f(-x) = -f(x)$ for all $x \in \text{Dom}(f)$ then the function $f(x)$ is said to be an odd function. Such functions are symmetric about the origin.

#Monotonicity of a Function

Monotonicity of a function is the property of a function whether it is increasing or decreasing.:

1. **Increasing function:** If for every $x_1, x_2 \in (a, b)$, $x_2 > x_1 \Rightarrow f(x_2) > f(x_1)$ then the function $y = f(x)$ is said to be increasing in the interval (a, b) .
or, If $x \uparrow \Rightarrow y \uparrow$ then function $y = f(x)$ is said to be increasing function.
2. **Decreasing function:** If for every $x_1, x_2 \in (a, b)$, $x_2 > x_1 \Rightarrow f(x_2) < f(x_1)$ then the function $y = f(x)$ is said to be decreasing in the interval (a, b) .
or, If $x \uparrow \Rightarrow y \downarrow$ then function $y = f(x)$ is said to be decreasing function.

#Transformation of Graphs

Shifting/Translation

Horizontal Shifting	$x \rightarrow x \pm a$	The graph of function $y = f(x - a)$, ($a > 0$) is the graph of $y = f(x)$ translated or shifted by a units to the right.
		The graph of function $y = f(x + a)$, ($a > 0$) is the graph of $y = f(x)$ translated or shifted by a units to the left.
Vertical Shifting	$y \rightarrow y \pm b$	The graph of function $y = f(x) + b$, ($a > 0$) is the graph of $y = f(x)$ translated or shifted by b units to the upward.
		The graph of function $y = f(x) - b$, ($a > 0$) is the graph of $y = f(x)$ translated or shifted by b units to the downward.

Reflection

Reflection about x-axis	$y \rightarrow -y$	The graph of function $y = -f(x)$ is the reflection of graph of $y = f(x)$ about the x-axis.
Reflection about y-axis	$x \rightarrow -x$	The graph of function $y = f(-x)$ is the reflection of graph of $y = f(x)$ about the y-axis.

#Sketching of Graph

Quadratic function

General Quadratic Function

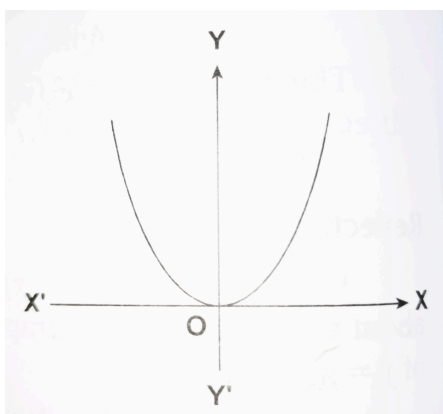
$y = f(x) = ax^2 + bx + c$ where a, b, c are constants and $a \neq 0$

Characteristics

1. The function represents a parabola.
2. If $a > 0$ then the parabola opens upward and if $a < 0$ the parabola opens downward.
3. **Intercepts:**
 - a. Y-intercept is $(0, c)$
 - b. X-intercepts are $(x_1, 0)$ and $(x_2, 0)$ if x_1, x_2 are its two roots.
4. **Vertex:** $(\frac{-b}{2a}, \frac{4ac - b^2}{4a})$ or $(\frac{-b}{2a}, f(\frac{-b}{2a}))$
5. **Axis/Line of Symmetry:** $x = \frac{-b}{2a}$
6. Domain = $(-\infty, \infty)$
7. Range = $(\frac{4ac - b^2}{4a}, \infty)$ for $a > 0$
 $= (-\infty, \frac{4ac - b^2}{4a})$ for $a < 0$

Example

The graph of $y = x^2$



Cubic function

General Cubic Function

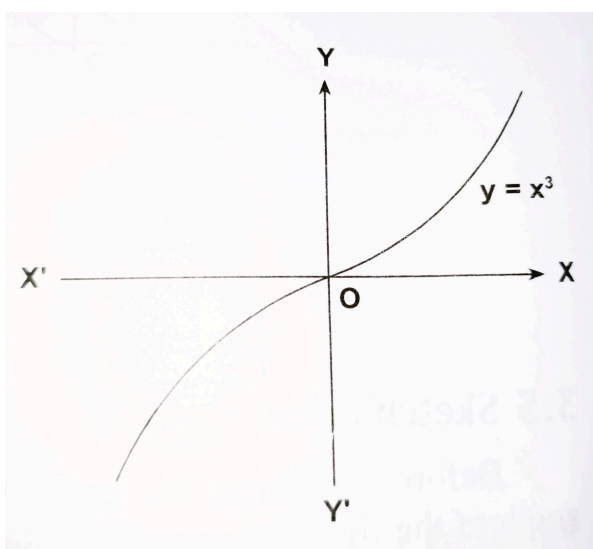
$y = f(x) = ax^3 + bx^2 + cx + d$ where a, b, c, d are constants and $a \neq 0$

Characteristics

1. The function represents a symmetric curve about the origin.
2. **Intercepts:**
 - a. Y-intercept is $(0, d)$
 - b. X-intercepts are $(x_1, 0)$, $(x_2, 0)$ and $(x_3, 0)$ if x_1, x_2, x_3 are its three roots.
3. Domain = $(-\infty, \infty)$
Range = $(-\infty, \infty)$

Example

The graph of $y = x^3$

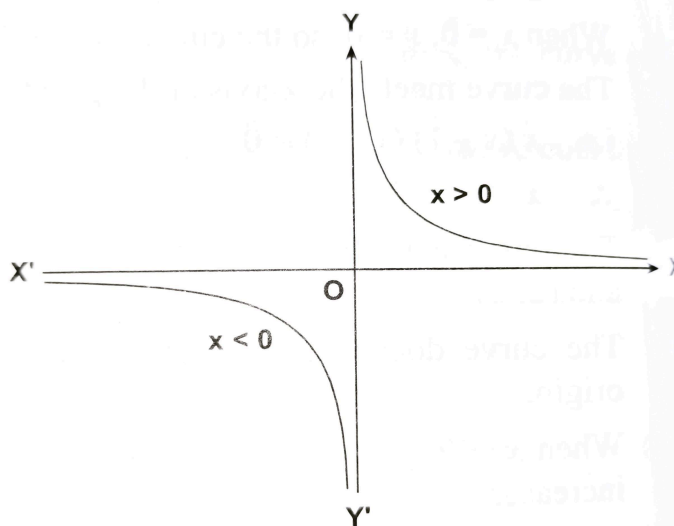


Reciprocal function

$$y = \frac{1}{x}$$

Characteristics

1. Both the axes are its asymptotes.
2. Domain = $(-\infty, \infty) - \{0\}$
Range = $(-\infty, \infty) - \{0\}$
3. The function is symmetric about the origin.
4. For $x > 0$ it lies in the first quadrant.
For $x < 0$ it lies in the third quadrant.



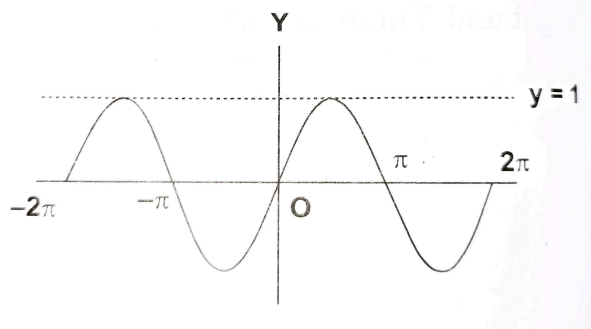
Trigonometric function

Sine Function

$$y = \sin x$$

Characteristics

1. When $x = 0$, $y = 0$ so the y-intercept is 0.
2. When $y = 0$, the x-intercepts are $\pm n\pi$, ($n \in \mathbb{N}$).
3. The curve is symmetric about the origin.
4. Its period is 2π .
5. Its maximum value is 1 and minimum value is -1.

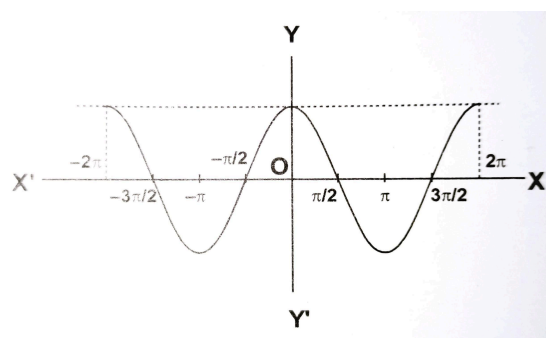


Cosine Function

$$y = \cos x$$

Characteristics

1. When $x = 0$, $y = 1$ so the y-intercept is 1.
2. When $y = 0$, the x-intercepts are $\pm \frac{(2n-1)\pi}{2}$, ($n \in \mathbb{N}$).
3. The curve is symmetric about the y-axis.
4. Its period is 2π .
5. Its maximum value is 1 and minimum value is -1.

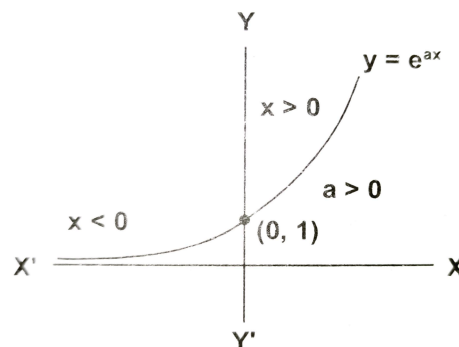


Exponential function

$$y = e^x$$

Characteristics

1. When $x = 0$, $y = 1$ so the y-intercept is 1.
2. The x-axis is its asymptote.
3. The curve is not symmetric.
4. Domain = $(-\infty, \infty)$
Range = $(0, \infty)$



Logarithmic Function

$$y = \log x$$

Characteristics

1. When $y = 0$, $x = 1$ so the x-intercept is 1.
2. The y-axis is its asymptote.
3. The curve is not symmetric.
4. Domain = $(0, \infty)$
Range = $(-\infty, \infty)$

